

Childhood Immunization

Contents

Childhood Immunization	1
General objectives	2
Introduction	2
Passive immunization	2
Passive natural immunization	2
Passive artificial immunization	2
Active immunization	2
Active natural immunization	3
Active artificial immunization	3
Vaccine and vaccination	3
Historical perspectives	3
Immunizing agents	3
Live attenuated vaccines	3
Inactivated vaccines	4
Immune globulins	4
Determinants of immune response	4
Classes of protective antibody	5
Vaccine recommendation	5
Objectives of the immunization programme	5
Vaccine delivery	5
Immunization in special circumstances	5
Contraindications	6
Precautions	6
NOT contraindications	6
Adverse reactions	6
Cost effectiveness	6
Immunization coverage	7
Missed opportunity	7
Cold chain	7
Cold chain equipment	7

Childhood Immunization

Dr Ijeoma Stella, Consultant Paediatrician – Paediatrics, 14/07/26

General objectives

- Define immunization
- Significance of immunization
- Types of immunity
- Immunizing agents
- National immunization schedule
- Contraindications to vaccinations
- Reactions to vaccines

Introduction

Immunization is defined as the **procedure by which the body is prepared to fight against specific disease.**

It is used to **induce the immune resistance of the body to a specific disease.**

Immunization is of two types:

- **Passive** immunization
- **Active** immunization

Passive immunization

Passive immunization is produced **without challenging the immune system of the body.**

It is produced by **administration of serum or gamma globulins from a person who is already immunized** (affected by the disease) **to a non-immune person.**

Passive immunization is acquired either **naturally or artificially.**

Passive natural immunization

Acquired from the **mother before and after birth.**

- **Before birth** – immunity is transferred from mother to fetus in the form of maternal antibodies (**mainly IgG**) **through the placenta**
- **After birth** – antibodies (**IgA**) are transferred through **breast milk**

Passive artificial immunization

Developed by **injecting previously preformed antibodies** using serum from humans or animals.

This type of immunity is useful for providing **immediate protection against acute infections** like **tetanus, measles, etc.**

Active immunization

Active immunization or immunity is acquired by **activating the immune system of the body.**

The body develops resistance against disease by **producing antibodies following exposure to antigens.** Active immunity is acquired either **naturally or artificially.**

Active natural immunization

Involves activation of the immune system in the body to produce antibodies. It is achieved in **both clinical and subclinical infections**.

Active artificial immunization

Achieved by the **administration of vaccines or toxoids**.

Vaccine and vaccination

Vaccine is a substance that is introduced into the body to prevent the disease produced by certain pathogens.

A vaccine consists of **dead pathogens** or **live but attenuated (artificially weakened) organisms**. The vaccine induces immunity against the pathogen, either by **production of antibodies** or by **activation of T-lymphocytes**.

Edward Jenner produced the first live vaccine. He produced the vaccine for **smallpox** from **cowpox virus**.

Vaccination is the **process of distributing and administering vaccine**.

Historical perspectives

- **Smallpox** – most feared; **20-30% case fatality rates**; leaving survivors **scarred for life**
- **Variolation** practised as early as the **10th century AD**
- **Edward Jenner** – first to publish, in **1796**
- **Rabies vaccine** by **Louis Pasteur**, 100 years later

Immunizing agents

VACCINE: any preparation which, when administered, is able to **stimulate the immune system** to produce specific responses that **inactivate, destroy or suppress** a given pathogen.

Two categories:

- **Live attenuated**
- **Inactivated**

Live attenuated vaccines

- Produced by **modifying a naturally occurring (wild) organism**, usually by **repeated culturing**
- **Retains the ability to replicate** and produce an immune response **similar to natural infection**
- **Circulating antibody may interfere** with the response
- **Long lasting immunity**
- **One dose usually suffices**

Examples – live attenuated viruses: poliomyelitis, measles, rubella, mumps, rabies, varicella, yellow fever, rotavirus, influenza

Examples – live attenuated bacteria: Bacille Calmette-Guérin (BCG), typhoid

Inactivated vaccines

- The organism is **inactivated with heat and/or chemicals (formalin)**. With **fractional** vaccines it is further treated to **purify only those components** to be included in the vaccine
- **Non-replicating**, so **cannot cause disease**
- **Less interference from circulating antibodies**
- **Requires more doses – 3-5**
- **Mostly humoral immune response**
- **Antibody titres diminish with time**

Examples – viral: poliomyelitis (Salk), hepatitis A, rabies, influenza

Examples – bacterial: pertussis, typhoid, cholera, plague, anthrax

Fractional:

- **Subunit** (made from a single protein) – hepatitis B, acellular pertussis, human papilloma virus
- **Toxoids** – diphtheria, tetanus
- **Polysaccharide** – pneumococcal, meningococcal, *Salmonella typhi*
- **Conjugate polysaccharide** – Hib, pneumococcal, meningococcal
- **Recombinant** (from recombinant DNA technology) – hepatitis B, human papilloma virus, influenza

Immune globulins

- **Immune globulin** – homologous pooled antibody, from many adults; contains antibodies to many antigens. Used for **hepatitis A, measles**
- **Homologous human hyperimmune globulin** – from humans with high levels of specific antibodies. Used for **post-exposure prophylaxis (PEP) of hepatitis B, rabies, tetanus, varicella**
- **Heterologous hyperimmune serum (antitoxin)** – from **animals**. Used for **botulism, diphtheria**
- **Monoclonal antibodies** – derived from a clone of antibody-producing cells. **RSV – palivizumab**

Determinants of immune response

Vaccine factors:

- Chemical and physical state of the vaccine
- Route/site of administration
- Vaccine dose
- Immunogenicity of the vaccine

Host factors:

- Age
- Sex
- Nutrition
- Pre-existing antibodies
- Genetics
- Immune status
- Pregnancy

Classes of protective antibody

- **Antitoxins** – inactivate soluble toxic protein products
- **Opsonins** – facilitate phagocytosis
- **Lysins** – interact with components of serum complement
- **Neutralising antibodies** – prevent proliferation of infectious virus
- **Antiadhesins** – prevent adhesion to mucosal surfaces

Vaccine recommendation

Factors to consider:

- Epidemiology – incidence, age, sex
- Age-specific morbidity and mortality
- Risk of adverse events
- Cost effectiveness
- Age of recommended health care visits
- Efficacy of the vaccine

Vaccinations are generally recommended at the age of:

- **Significant risk of disease and complications**
- **Protective immunologic response**

Also recommended for **special situations**, e.g. sickle cell anaemia (SCA), HIV.

Objectives of the immunization programme

- **Not just individual protection**
- **Decrease morbidity and mortality** from target diseases
- **Increase herd immunity** – the resistance of a group to invasion and spread of an infectious agent, due to immunity of a large proportion of the group
- **High coverage is needed** to achieve this

Vaccine delivery

- Routine immunization
- Response to an epidemic
- Supplemental immunization activities
- Mop up immunization
- Mobile or fixed locations

Immunization in special circumstances

Pneumococcal polysaccharide vaccine – SCA, anatomic asplenia, nephrotic syndrome, renal failure, HIV, organ transplant

HIV:

- **All vaccines if asymptomatic**
- **BCG and yellow fever contraindicated if symptomatic**
- **IPV preferred to OPV**

Also: cerebrospinal meningitis, rabies.

Live vaccines are generally not given to immunocompromised children.

Contraindications

- Anaphylaxis
- Encephalopathy
- Progressive neurologic disorders
- Immunodeficiency disorder; immunosuppressive therapy
- Pregnancy
- Any illness requiring admission to hospital – **defer** immunization

Precautions

- Fever $> 40.5^{\circ}\text{C}$
- Collapse / shock
- Inconsolable crying
- Guillain-Barré syndrome

For MMR:

- Blood transfusion
- Thrombocytopenia

NOT contraindications

- Upper respiratory tract infection
- Mild fever
- **Diarrhoea – repeat dose**
- Cerebral palsy
- Breastfeeding
- Antimicrobial therapy

Adverse reactions

- Pain, swelling, redness, soreness
- Fever
- Anaphylaxis
- Encephalopathy – convulsions, coma
- Rash / urticaria

Cost effectiveness

- Savings on treatment cost
- decreased mortality
- decreased morbidity and disability
- Avoidance of intangible costs (suffering)
- External / spillover benefits

Examples:

- **Smallpox eradication – saving 32 billion dollars**
- Poliomyelitis eradication
- Hib immunization

Immunization coverage

The **percentage of the target population that have received immunization.**

- Often evaluated in **children aged 12-23 months** – estimates recent immunization performance
- May be evaluated in the target population for a given vaccine, e.g. under-fives following NIDs

Missed opportunity

Occurs when an individual **in whom there is no contraindication does not receive the vaccine(s) for which he/she is eligible when he/she visits a health facility.** Can occur during curative or preventive visits.

Causes of missed opportunities:

- Non-availability of vaccine
- Not checking immune status
- Failure to give simultaneous injections
- Failure to provide info to parents
- Failure to immunize every day

Cold chain

The **cold chain is a system of storage and transport of vaccines at low temperatures, by the manufacturer to the actual vaccination site.**

The cold chain system is necessary because **vaccine failure may occur due to failure of storage and transport under strict temperature controls.**

- Vaccines stored in the **freezer compartment: polio and measles**
- Vaccines which must be stored in the **cold compartment and never allowed to freeze: BCG, DPT, DT, TT, typhoid, and diluents**
- Vaccines must be **protected from sunlight** and prevented from **contact with antiseptics**

At health centres, most vaccines (**except polio**) can be stored up to **5 weeks**, if the refrigerator temperature is strictly kept between **4 °C and 8 °C.**

Storage of opened vial:

- **With preservatives – up to 3 hours**
- **Without preservatives – up to 1 hour**

Cold chain equipment

- Walk-in cold rooms
- Deep freezer
- Small deep freezer
- Cold boxes
- Vaccine carrier
- Day carrier
- Ice packs